

IIoT Network Analysis: Age of Information and Reliability Trade-offs

In this lab, the main goal was to understand how Age of Information (AoI) and network reliability are related in Industrial Internet of Things (IIoT) systems. AoI basically measures how fresh the data is, while reliability refers to how successfully data is delivered, usually measured with packet loss probability (PLP). Throughout the lab, we explored how different network settings affect these two factors and how they can sometimes conflict with each other. This idea connects directly to the reading by Farag, Ali, and Stefanović (2023), which explains that improving reliability can sometimes make AoI worse, and vice versa.

At the beginning of the lab, we explored the dataset using pandas. The dataset included variables like transmission probability, capture threshold, number of nodes, channel quality, AoI, and packet loss probability. When looking at the data, one of the first things I noticed was that AoI had some really large values and outliers. These outliers made the graphs look weird at first, so we had to filter them out to better understand the overall trends. This step helped show that most of the time the network keeps data fairly fresh, but there are occasional spikes where delays become very large.

After cleaning the data, we created visualizations to better understand the relationships between variables. The histogram showed that most AoI values are on the lower end, which means the system usually works well. However, the presence of outliers shows that delays can still happen under certain conditions. The scatter plot didn't show a strong linear relationship between AoI and transmission probability, but it did show that network performance changes depending on how the system is set up. This suggests that AoI is affected by multiple factors, not just one.

Next, we built a Random Forest model to try to predict Aol based on the other network features. After running the model, the results showed a high Mean Squared Error (MSE) and a negative R^2 value. This basically means the model didn't do a good job predicting Aol. At first, that seemed like a problem, but it actually makes sense because Aol is very inconsistent and influenced by sudden changes in the network. Even though the model didn't perform well overall, the feature importance results were still helpful. They showed that capture threshold and number of nodes had the biggest impact on Aol, followed by channel quality and transmission probability.

For the bonus part, we created a simple neural network to predict both Aol and packet loss probability at the same time. The results were mixed. The model still struggled with predicting Aol, but it did a much better job predicting PLP. This suggests that PLP is easier to predict because it's more stable, while Aol is more unpredictable. Comparing the two models, the Random Forest was more reliable for Aol, while the neural network handled PLP better.

These results connect well with the reading by Farag et al. (2023). The article explains that there is a trade-off between Aol and reliability in IIoT networks. For example, if you increase transmission probability, you can reduce packet loss and improve reliability, but it can also cause more congestion, which increases Aol. On the other hand, reducing transmissions might improve Aol but make reliability worse. This trade-off showed up in the lab results, where different settings affected Aol and PLP in different ways.

Another important point from the reading is that IIoT networks are complex, especially when different types of devices and conditions are involved. This makes it hard for machine learning models to predict things like Aol accurately. The fact that our Random Forest model didn't perform well supports this idea, since it struggled to capture all the variability in the data.

Overall, this lab showed that balancing data freshness and reliability is not simple. You can't just optimize one without affecting the other. Even though the models weren't perfect, they helped identify which factors matter the most and gave a better understanding of how IIoT networks behave. These insights are important for real-world applications like smart factories, healthcare monitoring, and environmental systems, where having accurate and up-to-date data is critical.

In conclusion, this lab helped reinforce the main idea from the reading, which is that there is always a trade-off between AoI and reliability in IIoT systems. By working with real data and building models, it became clear that optimizing these systems requires finding the right balance instead of focusing on just one goal.